

BRIAN ANTHONY EPSTEIN

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Education

Georgia Institute of Technology

Atlanta, Georgia

Ph.D. Mechanical Engineering, GPA: 4.00 | Research Area: Autonomy and Controls

August 2022 – Fall 2026

M.S. Mechanical Engineering, GPA: 4.00 | Controls & Robotics, Mechatronics & Design

August 2022 – May 2024

B.S. Mechanical Engineering, GPA: 3.96 | Minor: Robotics, Honors Program, Highest Honors

August 2018 – May 2022

Research Experience

Georgia Tech Aerial Robotics and Experimental Autonomy Lab (AREAL)

August 2022 – Present

Graduate Researcher (PI: Dr. Jonathan Rogers)

Atlanta, Georgia

Solar-Powered Autonomous Seaplane (Dissertation)

August 2023 – Present

- Leading end-to-end design, build, and flight test of a 5.0 kg solar-powered uncrewed seaplane UAV.
- Generated VSPAero drag polars, lift curves, and stability derivatives that drove PX4 control tuning.
- Invented a spanwise rotating wing for solar tracking and self-righting, simulated $\sim 22\%$ gain over a fixed array.
- Sized the 3D-printed hull to metacentric height targets, verified statically stable moored in open water.
- Fabricated airframes via additive manufacturing in ASA with a carbon-fiber spar and fiberglass-epoxy hull layup.
- Established a rapid-prototyping print-and-composite process yielding a complete new airframe in ~ 3 days.
- Waterproofed the airframe for open-water landings with conformal coating, potting, and gasketed hatches.
- Implemented PX4 modules for onboard solar tracking with encoder feedback and autonomous self-righting.
- Authored a custom JSBSim airframe model and ran PX4 SITL in Gazebo before every flight-software change.
- Executing flight tests from runway and lake surface, validating takeoff run, glide, and 150 W cruise.
- Developing a control barrier function safety filter for open-water landing with impact and stall limits.
- Developed energy-aware fly-vs-float mission autonomy for the seaplane under stochastic wind and solar.
- Solved the policy as a finite-horizon MDP by stochastic dynamic programming, cutting mission failure rate 30%.
- Ran sensitivity studies across region, season, failure penalty, and mission duration, confirming policy robustness.
- Cut required battery capacity from 450 Wh to 250 Wh versus tuned threshold baseline controllers.

University Consortium for Applied Hypersonics (Fault-Tolerant Estimation)

August 2025 – June 2026

- Developed state estimation for fault-tolerant control of hypersonic vehicles in Simulink and C++.
- Analyzed hypersonic flight test data to characterize vehicle behavior and inform estimator design.

Pseudo-Cyclic Contra-Rotating Helicopter (Ground Test and Actuator Validation)

May 2023 – August 2023

- Developed a custom PX4 actuator driver in C++ with uORB messaging for a pseudo-cyclic control mechanism.
- Supported a 4-person ground-test campaign with LabView data acquisition, validating target thrust.
- Ground tests exposed insufficient control bandwidth and hysteresis, informing the decision to halt development.

Autonomous Drone Capture (Quadcopter Interceptor) | Sole Developer

August 2022 – May 2023

- Designed and built a 500 mm quadcopter with a capture basket to catch a target drone mid-air under Vicon tracking.
- Built a Python pursuit-intercept guidance stack driving nested PID loops over a PX4-ROS2 offboard link.

Autonomous Landing on a Moving Platform (Undergraduate Research)

January 2022 – May 2022

- Developed C++ device drivers and a micro-ROS link to ROS2 nodes for gimbal control on the landing UAV.

Publications

Epstein, B., and Rogers, J. D., “Optimal Flight Planning for Marine Mammal Surveillance from a Seaplane UAV,” *Journal of Aircraft* (in press).

Epstein, B., Trujillo, N., Mogge, A., and Rogers, J. D., “Design and Flight Demonstration of a 3D-Printed Solar-Powered Seaplane UAV for Persistent Marine Mammal Monitoring,” *AIAA SciTech Forum*, 2027 (in press).

Epstein, B., and Rogers, J. D., “Conceptual Design of a Low-Altitude Solar Powered Seaplane for Oceanic Monitoring,” *AIAA AVIATION Forum and ASCEND*, AIAA Paper 2024-3751, 2024.

Ward, D., **Epstein, B.**, Tiziani, L., and Hammond, F. L., III, “Optimal Design of a Mechatronic Lever Arm for Pneumatic Exoskeleton: Design and Validation,” *Proceedings of the Design of Medical Devices Conference*, 2021.

Projects

AI Grand Prix: Autonomous Drone Racing | *Team AI of Sauron* May 2026 – Present

- Developing GPS-denied autonomy on classical controls and estimation that flies a racing drone with no pilot.
- Developed a 15-state error-state Kalman filter propagating IMU at ~115 Hz with monocular gate updates.
- Localized gates in OpenCV with red-dominance segmentation, quad fitting, and PnP for 6-DOF gate pose.
- Modeled gate sightings with anisotropic noise so monocular range error cannot corrupt across-track state.
- Generated an arc-length Hermite spline through the gate openings with a curvature-bounded speed governor.
- Proposed and tuned the team's SE(3) geometric tracking controller with curvature feed-forward.
- Flew the 17-gate, 253 m virtual qualifier course in 17.6 s from camera and IMU alone at 14.4 m/s mean.
- Ranked 7th of 3,000+ teams worldwide in the Anduril-backed race and qualified for the hardware round.

Combat Robotics: Chonkiv | *Georgia Tech Competitive Robotics Club, RoboJackets* August 2018 – Present

- Won 2 NHRL championships and the first collegiate featherweight title, ranked 1st in the world for a year.
- Packaged motors, ESCs, battery, and wiring harness in a sealed high-shock volume with strain relief.
- Modularized drive pods and weapon assembly for hand-tool field repair inside NHRL's match turnaround.

Quadcopter Double Slung Load Input Shaping Spring 2024

- Implemented input-shaped trajectories with a custom controller over the PX4-ROS2 bridge on a quadcopter.
- Reproduced the quadcopter and slung load in Gazebo SITL, validating simulation against real-world tests.

Work Experience

Georgia Institute of Technology Spring 2024 and Spring 2025

AE4803 Flight Test Engineering Course Assistant *Atlanta, Georgia*

- Developed Simulink flight-dynamics models and integrated the backend with X-Plane 12 for real-time state.
- Co-designed flight-test lab assignments on the AE simulator suite in the course's first hands-on FTE term.

JLG Industries May 2022 – August 2022

Engineering Research and Development Intern *Hagerstown, Maryland*

- Built vision-based depth perception and plane detection with RANSAC, Open3D, and OpenCV on RealSense.
- Prototyped an autonomous powerwashing demo on a 6-DOF robotic manipulator with JLG hardware and ROS2.

Eaton Corporation May 2021 – August 2021

Robotic Manufacturing Intern, Operations Leadership Development Program *Hastings, Nebraska*

- Programmed a Fanuc robot and CNC lathe to automate gear machining rework, cutting cycle time 75%.
- Built a cycle-time-based bottleneck prioritization system for gear machining, saving \$40,000.

East West Manufacturing January 2020 – May 2020

Engineering Intern *Atlanta, Georgia*

- Designed prototype eddy-current brake components, authored GD&T drawings, and fabricated the parts.
- Built a vacuum thermoforming table and a hemi-anechoic chamber for HVAC-motor acoustic testing.

Lockheed Martin May 2019 – August 2019

Manufacturing Engineering Intern *Ocala, Florida*

- Reduced missile assembly defect rate with an Arduino-controlled smart fixture, built \$16,000 under quote.
- Integrated the smart fixture with the existing torque driver system, minimizing defect escapes.

Skills and Coursework

Software: Python, C++, MATLAB, Simulink, ROS2, PX4, MAVLink, Gazebo, JSBSim, X-Plane, OpenCV, VSPAero

Hardware: 3D printing, composites, CNC, wiring harnesses, environmental sealing, payload integration

Coursework: Advanced Control System Design and Implementation, Linear and Nonlinear Control Systems

Leadership

Woodruff Graduate Leaders Fellow | *Woodruff School of Mechanical Engineering* 2025

- Selected to the inaugural 22-fellow cohort of a College of Engineering graduate leadership program.
- Represented graduate student concerns to administration, proposing solutions adopted into the program.

Undergraduate Research Mentor | *Aerial Robotics and Experimental Autonomy Lab* 2023 – Present

- Mentored over 25 undergraduate researchers across design, manufacturing, controls, and flight testing.

RoboJackets Combat Robotics | *Georgia Tech Competitive Robotics Club* August 2018 – Present

- Held Mechanical Training Lead, Sponsorship Chair, Team Lead, and Team Mentor roles across 8 years.
- Serve on the board of RoboJackets Inc., the nonprofit managing sponsorship funds and competition winnings.